

JIGNA KALSARIYA

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EDUCATION

Sardar Vallabhbhai National Institute of Technology, Surat

Bachelor of Technology in Computer Science

Surat, India

July 2024 – July 2028

- CGPA: 8.64

PROJECTS

CardioVision | JavaScript, MediaPipe, Computer Vision, rPPG, Signal Processing

GitHub | Website

- Engineered a browser-based remote photoplethysmography (rPPG) system capable of estimating heart rate in real time using only a standard webcam, eliminating the need for wearable sensors.
- Implemented MediaPipe Face Mesh to track 468+ facial landmarks and dynamically extract multi-region facial signals from the forehead and cheeks for robust pulse detection.
- Developed a custom signal-processing pipeline incorporating detrending, CHROM-based pulse extraction, Hamming windowing, and Fourier Transform analysis to isolate physiological heart-rate frequencies.
- Designed a real-time cardiovascular assessment dashboard with BPM visualization, heart-rate classification (Bradycardia, Normal, Tachycardia), and configurable clinical measurement protocols.
- Built session reporting and JSON export functionality, enabling structured storage and analysis of cardiovascular measurement data.
- Optimized the signal-processing pipeline for real-time performance, reducing computation overhead while maintaining accurate heart-rate estimation.

WanderLust | Node.js, Express.js, MongoDB, EJS, Passport.js, Cloudinary, REST API

GitHub | Website

- Developed a full-stack property rental platform with CRUD functionality for listings, reviews, and user management using Node.js, Express.js, and MongoDB.
- Implemented session-based authentication and authorization using Passport.js, express-session, and MongoStore with protected and ownership-based routes.
- Designed and integrated 15+ RESTful API endpoints using modular Express routers for listings, users, authentication, and reviews.
- Integrated Multer and Cloudinary for property image uploads, cloud storage, and efficient media management.
- Implemented Joi validation, centralized error handling, and Mongoose references/population for reliable data management.

TECHNICAL SKILLS

Programming Languages: C++, C, Python, JavaScript, SQL

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS, Bootstrap

Databases: MySQL, MongoDB

Developer Tools: Git, GitHub

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks

ACHIEVEMENTS & CERTIFICATIONS

Google Cloud Arcade Facilitator Program 2025

Aug 2025

- Achieved the Ultimate Milestone in the Google Cloud Arcade Facilitator Program (Cohort 1).
- Gained hands-on experience with Google Cloud technologies and contributed to the learning community.

Finalist – Web Wonders Competition (SVNIT Surat)

July 2025

- Led a team as Team Leader and secured a finalist position in the Web Wonders competition organized by SVNIT.
- Coordinated team efforts and contributed to developing a web-based solution on the Transportation theme.

LEADERSHIP & ACTIVITIES

- **Youth Motivational Speaker** – Delivered motivational sessions for students on self-confidence, goal setting, and personal development, fostering a positive and growth-oriented mindset.
- **Participant, Viksit Bharat Young Leaders Dialogue (VBLYD'26)** – Selected among 50 lakh+ participants for a national-level youth leadership platform; contributed ideas on policy innovation, youth empowerment, and nation-building aligned with the vision of Viksit Bharat 2047.